INCH-POUND

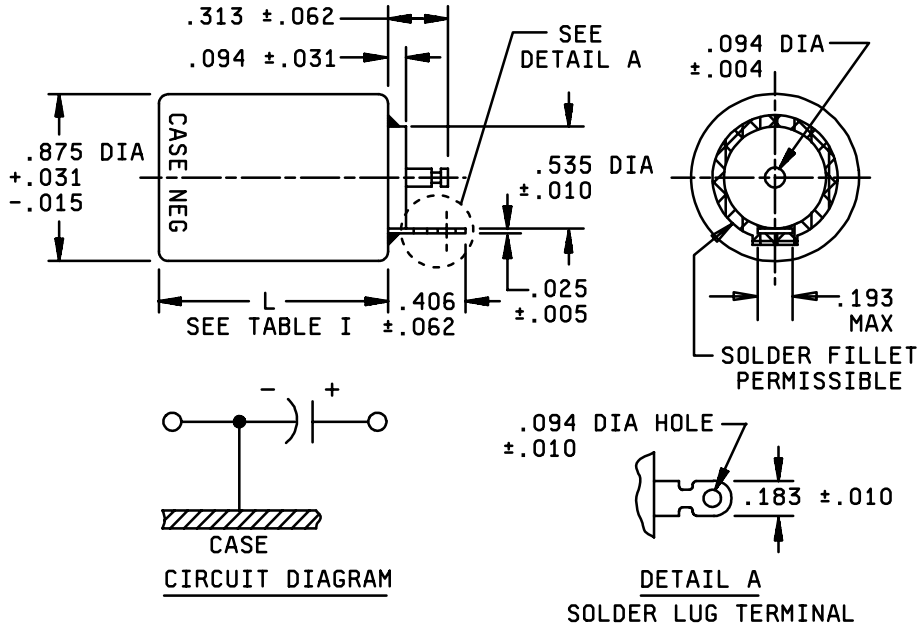
MIL-PRF-39006/19A
6 August 2001
SUPERSEDING
MIL-C-39006/19
28 August 1970

MILITARY SPECIFICATION SHEET
CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM,
(POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C),
ESTABLISHED RELIABILITY,
STYLE CLR14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39006.

INACTIVE FOR NEW DESIGN AS OF
23 JUNE 1981.
FOR NEW DESIGN USE MIL-PRF-39006/22.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .004 | 0.10 | .094 | 2.39 |
| .005 | 0.13 | .183 | 4.65 |
| .010 | 0.25 | .193 | 4.90 |
| .015 | 0.38 | .313 | 7.95 |
| .025 | 0.64 | .406 | 10.31 |
| .031 | 0.79 | .535 | 13.59 |
| .062 | 1.57 | .875 | 22.23 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style CLR14 capacitors.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

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TABLE L. Capacitor characteristics.

| Rated voltage (85°C) | Cap. | DC leakage (max) | | | Dissi-pation factor (max) | Derated voltage (125°C) | Surge voltage (85°C) | Imped-ance (max) | Dimension L ±.062 (1.57) | Part No. M39006/19- | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Failure rate level (% 1,000 hr) | | | | |
| | | 25°C | 85°C | 125°C | | | | | | L(2.0) | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| volts, dc | µF | µA | µA | µA | percent | volts, dc | volts, dc | ohms | Inches | | | | | |
| 20 | 100.0 | 10 | 50 | 80 | 21.0 | 18 | 23 | 30 | 0.540 (13.72) | 1000 | 1020 | 1040 | 1060 | 1080 |
| 20 | 200.0 | 16 | 80 | 128 | 36.0 | 18 | 23 | 20 | 0.732 (18.59) | 1001 | 1021 | 1041 | 1061 | 1081 |
| 30 | 75.0 | 11 | 55 | 88 | 15.0 | 26 | 34.5 | 45 | 0.540 (13.72) | 1002 | 1022 | 1042 | 1062 | 1082 |
| 30 | 150.0 | 13 | 90 | 104 | 29.0 | 26 | 34.5 | 30 | 0.732 (18.59) | 1003 | 1023 | 1043 | 1063 | 1083 |
| 60 | 40.0 | 12 | 60 | 96 | 8.2 | 52 | 69 | 65 | 0.540 (13.72) | 1004 | 1024 | 1044 | 1064 | 1084 |
| 60 | 80.0 | 19 | 95 | 152 | 16.0 | 52 | 69 | 35 | 0.732 (18.59) | 1005 | 1025 | 1045 | 1065 | 1085 |
| 90 | 25.0 | 11 | 55 | 88 | 5.1 | 78 | 103 | 90 | 0.540 (13.72) | 1006 | 1026 | 1046 | 1066 | 1086 |
| 90 | 50.0 | 18 | 90 | 144 | 10.0 | 78 | 103 | 45 | 0.732 (18.59) | 1007 | 1027 | 1047 | 1067 | 1087 |
| 180 | 12.0 | 11 | 55 | 88 | 5.1 | 155 | 207 | 180 | 0.920 (23.37) | 1008 | 1028 | 1048 | 1068 | 1088 |
| 180 | 25.0 | 18 | 90 | 144 | 10.0 | 155 | 207 | 90 | 1.300 (33.02) | 1009 | 1029 | 1049 | 1069 | 1089 |
| 270 | 8.0 | 11 | 55 | 88 | 5.1 | 235 | 310 | 270 | 1.270 (32.36) | 1010 | 1030 | 1050 | 1070 | 1090 |
| 270 | 16.0 | 18 | 90 | 144 | 10.0 | 235 | 310 | 135 | 1.865 (47.37) | 1011 | 1031 | 1051 | 1071 | 1091 |
| 360 | 6.0 | 11 | 55 | 88 | 5.0 | 310 | 414 | 360 | 1.635 (41.53) | 1012 | 1032 | 1052 | 1072 | 1092 |
| 360 | 12.0 | 18 | 90 | 144 | 10.0 | 310 | 414 | 180 | 2.420 (61.47) | 1013 | 1033 | 1053 | 1073 | 1093 |
| 450 | 5.0 | 11 | 55 | 88 | 4.9 | 390 | 518 | 450 | 2.000 (50.80) | 1014 | 1034 | 1054 | 1074 | 1094 |
| 450 | 10.0 | 18 | 90 | 144 | 9.8 | 390 | 518 | 225 | 2.980 (75.69) | 1015 | 1035 | 1055 | 1075 | 1095 |
| 540 | 4.0 | 11 | 55 | 88 | 5.1 | 470 | 621 | 540 | 2.365 (60.07) | 1016 | 1036 | 1056 | 1076 | 1096 |
| 540 | 8.0 | 18 | 90 | 144 | 10.0 | 470 | 621 | 270 | 3.532 (89.71) | 1017 | 1037 | 1057 | 1077 | 1097 |
| 630 | 3.5 | 11 | 55 | 88 | 5.0 | 545 | 724 | 630 | 2.720 (69.09) | 1018 | 1038 | 1058 | 1078 | 1098 |
| 630 | 7.0 | 18 | 90 | 144 | 10.0 | 545 | 724 | 315 | 4.062 (103.17) | 1019 | 1039 | 1059 | 1079 | 1099 |



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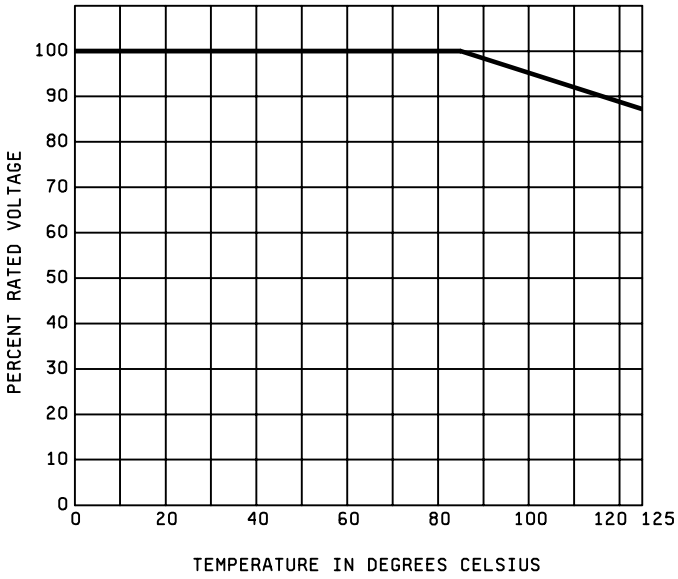


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Case type: Tubular, uninsulated.
Seal type: G (hermetic).
Terminals: Solder-lug.
Rated temperature: -55° to +85°C, voltage derated to +125°C (see figure 2).
Rated voltage: See table I.
Capacitance (Cap.) (nom): See table I.
Cap. tolerance: +75, -15 percent.
FR level: See table I.

Constant voltage conditioning: As specified in MIL-PRF-39006.

DC leakage (DCL) (max) (at 25°C): See table I.

Cap. (nom): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (max): See table I.

Seal: Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).

Thermal shock: Method 107 of MIL-STD-202, condition A (with step 3 at +125°C). Number of cycles - 30.
DCL: 125 percent (max) of 25°C value of table I.
$\Delta$Cap.: Within $\pm$5 percent of initial measurement.
DF: 115 percent (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hr).



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Solderability: Not applicable.

Terminal strength:
Pull: Method 211 of MIL-STD-202, condition A.
Bend (if required): Method 211 of MIL-STD-202, condition B.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.
DCL: 125 percent (max) of 25°C value of table I.
ΔCap.: Within ±8 percent of initial measurement.
DF: 115 percent (max) of initial requirement.

Dielectric withstanding voltage (DWV): Not applicable.

Insulation resistance (IR): Not applicable.

Low temperature (storage): Method 502 of MIL-STD-810.
DCL: See table I.
ΔCap.: Within ±5 percent of initial measurement.
DF: See table I.

Stability at low and high temperatures:
Step 1 (+25°C):
DCL: See table I.
Cap.: Within tolerance of table I value.
DF: See table I.
Step 2 (-55°C):
Imp.: See table I.
ΔCap.: -60 percent (20 V), -45 percent (30 V), or -35 percent (>30 V) of step 1 value (max).
Step 3 (+25°C):
DCL: See table I.
ΔCap.: Within ±5 percent of step 1 value.
DF: See table I.
Step 4 (+85°C):
DCL: See table I.
ΔCap.: +15 percent (<30 V) or +10 percent (>30 V) of step 1 value (max).
DF: See table I.
Step 5 (+125°C): 1/
DCL: See table I.
ΔCap.: +20 percent (<26 V) or +10 percent (>26 V) of step 1 value (max).
DF: See table I.
Step 6 (+25°C):
DCL: See table I.
ΔCap.: Within ±5 percent of step 1 value.
DF: See table I.

Reverse voltage: Not applicable.

Life at 85°C: Method 108 of MIL-STD-202.
2,000-hr (qualification):
DCL: See table I.
ΔCap.: Within ±15 percent of initial measurement.
DF: See table I.

1/ 125°C derated voltage referenced.



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10,000-hr (group B) and extended life:
DCL (85°C): 125 percent (max) of table I value.
DCL (25°C): See table I.
ΔCap.: Within ±25 percent of initial measurement.
DF: 200 percent (max) of table I value.
DWV and IR: Not applicable.

Vacuum life: Not applicable.

Life at 125°C: Method 108 of MIL-STD-202.
DCL: See table I.
ΔCap.: Within ±15 percent of initial measurement.
DF: 130 percent (max) of table I value.
DWV and IR: Not applicable.

10,000-hr (group B):
DCL (125°C): 125 percent (max) of table I value.
DCL (25°C): See table I.
ΔCap.: Within ±25 percent of initial measurement.
DF: 200 percent (max) of table I value.
DWV and IR: Not applicable.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 feet).
Test voltage - Rated voltage for 20 to 450 V units; 500 V dc for 540 and 630 V units.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
DCL: See table I.
ΔCap.: Within ±5 percent of initial measurement.
DF: See table I.

Marking: In accordance with MIL-C-39006.
Polarity - Capacitors shall be marked “CASE NEG” on the negative end (see figure 1).

Voltage groups (85°C rated voltage):

| | | |
|---|---|---|
| I | ..... | 20 to 90 V inclusive |
| II | ..... | 180 to 360 V inclusive |
| III | ..... | 450 to 630 V inclusive |

Custodians:
Army - CR
Navy - EC
Air Force - 11

Preparing activity:
DLA - CC

(Project 5910-2136)

Review activities:
Navy - AS, MC, OS
Air Force - 19



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